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Optional subtitle is on a separate line

by

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DOCTOR OF PHILOSOPHY
(POPULATION, HEALTH AND PLACE)

May 2026

To a very important person

Acknowledgements

This is an optional acknowledgment section. It should be numbered. Note the `\noindent` command for the first paragraph after the heading (and this is true for all text that follows headings headings after this).

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Abbreviations (Optional)

ANC	Antenatal care
CrI	Credible interval
DHS	Demographic and Health Survey
EHP	Essential Health Package
EPSeM	Equal probability of selection method
HSE	Health service environment
IMCI	Integrated Management of Childhood Illness
LMIC	Low- and middle-income country
NGO	Nongovernmental organization
OR	Odds ratio
PD	Posterior probability of direction
PP	Posterior probability
SARA	Service Availability and Readiness Assessment
SC	Sick-child care
SEA	Standard enumeration area
SPA	Service Provision Assessment
SRI	Service Readiness Index
UHC	Universal health coverage
USAID	United States Agency for International Development
WHO	World Health Organization

Abstract

As of 2026, the Thesis Center requires an abstract of NO MORE than 250 words. This can be quite limiting, but I would encourage you to write this early this and your title is *required* before anything on your thesis center profile can be saved. There does not seem to be a clear limit on the abstract length in the actual text.

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Chapter 1 Introduction

Again, note the `\noindent` command to start this section off. You should also use the `\autocite` command for citations, rather than the normal "cite" command for the package that supports Chicago Manual of Style (Rozelle et al. 2023). Chapter headings use the `\chapter{Chapter Name}` command.

Anything you reference here will be automatically added to your references at the end. You can also use a `\citeyear` command to cite the year, as Rozelle, et al. (2023) sometimes does. This will add it to the bibliography, but be sure to add in parentheses, as these are not automatically added with the `\citeyear` command. You can

1.1 Section

Here is a section (using the `\section` command), for a level one heading. I'm going to write a bit more text so this does not look weird.

I would also note that when you use one of these commands, (and this also applies to figures, tables, and equations) you can establish a label to reference that item dynamically using the `\label{labelname}` command. Then you may reference it using the `\ref{label-name}` command (for example, see Section 1.2).

I like to use prefixes like `ch1:` and `tab:` or `fig:` to help me remember what I'm referencing (in $\text{\LaTeX}$ using overleaf, these show up as suggestions for autofill; see Section 1.1.1). I can also reference a figure or section heading in another chapter (see Table 2.1).

1.1.1 Level 2 heading

Here is a section (using the `\subsection{Level 2 heading title}` command), for a level two heading. Subsections still appear on the in the Table of Contents.

1.1.1.1 Level 3 heading

If absolutely necessary, you can use a level 3 heading with with the `\subsubsection{Level 3 heading title}` command), for a level three heading. Level 3 headings (or subsubsections) do not appear in the in the Table of Contents.

1.2 Another section

Other quick notes, the PHP formatting guide requires left alignment with a ragged right edge. This and spacing should generally be handled by this template.

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Chapter 2 Example Chapter Two (And Some Notes About Reference Management)

You will find some additional examples of how references are formatted in the `main.tex` file (which organizes all the pieces of the $\text{\LaTeX}$ document like chapters, titles, and applies some additional formatting). In the preamble, observe the `\addbibresource{betterbiblatex-references.bib}`. This is what tells $\text{\LaTeX}$ where to look for references.

You can automatically link reference managers like Zotero and Mendeley to Overleaf. However, there were some small formatting things I wished to edit that this integration did not solve. Specifically, using short names. For example, "World Health Organization" is better presented as "WHO" in-text, but I want the full "World Health Organization" in the reference list. The default integration with overleaf does not accommodate that.

Instead, it can be accomplished with the Better BibTex plugin on Zotero and an added field (i.e., adding `"tex.shortauthor: WHO"` to the "Extra" field in Zotero), then exporting the full library with Better BibTex. You can then upload this export manually to Overleaf. A small extra step in the workflow, but I did not find it too annoying, since my reference library was not frequently updated at the dissertation assembly stage.

2.1 Methods

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2.2 Results

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Table 2.1: Service Provision Assessment sample for Tanzania, Nepal, Malawi, Haiti, and Afghanistan

	Health Facilities			Client visits observed	
	Sampling frame	Sampled	Successfully audited	ANC	Sick child
Afghanistan 2018–19	231	160	142	491	574
Haiti 2017–2018	1,033	1,033	1,007	1,526	2,166
Malawi 2014	1,060	1,060	977	2,105	3,329
Nepal 2015	4,719	992	963	1,509	2,186
Tanzania 2014–15	7,102	1,200	1,188	4,007	4,961

Note: ANC = antenatal care.

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2.3 Discussion

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Chapter 3 Example Chapter Three: Some Notes About Table and Figure Formatting

Tables and figures are a bit annoying. Especially in keeping with the formatting guidelines.

Remember that table and figure text can be no smaller than font size 9, and no larger than 12. This means I re-rendered many of my figures in R to meet this criteria. Tables, at times, needed to be split or compressed to fit within the margins.

The figures and tables below are from another paper of mine that were published (Rozelle 2026a). I would flag that LANDSCAPE ORIENTED TABLES MUST BE RENDERED IN PORTRAIT MODE. This is something that caught a few of the PhD candidates I spoke to, including myself. The current $\text{\LaTeX}$ template manages this the way the graduate school wants it, but uses a less standard package to do so.

ChatGPT or Claude are great at helping to format tables and getting them to fit in a page until you can do it yourself, but I would recommend using the `threeparttable` package. It helps keep everything tidy, from caption to notes (see 3.1). Tables and figures in the appendix come from my dissertation (Rozelle 2026b).

3.1 Methods

3.1.1 Analytic approach

Look, $\text{\LaTeX}$ can do some cool formulas (one of my favorite things about it (see Equation 3.1)! For an even more interesting equation, check Appendix A.

$$y_{ik} = \beta_0 + \beta_1 \text{Fees}_i + \beta_2 \text{Rural}_i + \beta_3 \mathbf{F}_i + \beta_4 \text{Services}_i + u_{g(k)} + \epsilon_{ik} \quad (3.1)$$

In the equation, y_{ik} is the example outcome for stratum k . In rhoncus, ante ornare laoreet dictum, nunc diam laoreet risus, sed elementum lacus tellus nec sapien. Integer malesuada ante at turpis pulvinar feugiat. Aenean fermentum tincidunt molestie. Integer pharetra mattis tortor vel auctor. Aenean finibus a felis sit amet rhoncus. Phasellus non lectus et enim aliquet fermentum et ut nunc. Curabitur tincidunt placerat gravida.

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All facility-level regression coefficients (including the client-fee indicator and covariates) were assigned improper flat priors over $\mathbb{R}$. The model intercept was assigned a Student-t prior with three degrees of freedom, centered at the observed mean of the outcome. The standard deviation of the managing authority by facility-type random intercept, as well as the residual standard deviation, were assigned half Student-t priors with three degrees of freedom and scale 2.5 (Student-t(3, 0, 2.5), constrained to be non-negative).

All models were fit with cmdstanr via brms using 8,000 iterations (4,000 warmup) per chain across eight chains. Convergence was assessed using standard diagnostics (Rhat = 1.00 and effective sample sizes), alongside posterior predictive checks. All monitored parameters met these convergence thresholds (Rhat = 1.00).

3.2 Results

Look below at the landscape table here, which is a great table (see Table 3.1). If you write a dissertation, I hope your tables look this good.

Table 3.1: This is a much longer caption, and I can include a few more details here if I want, or different text than what is in the TOC altogether

	Afghanistan (2018)	Bangladesh (2017)	Ethiopia (2021–22)	Malawi (2013–14)	Nepal (2021)	Tanzania (2014–15)
Number of facilities (<i>n</i>)	92	1,338	1,060	782	1,297	928
Facilities charging client fees (%)	75.0	37.2	78.8	38.5	30.9	91.8
Overall SRI (0–10)	7.70	4.56	6.23	6.00	5.90	5.53
Basic amenities	8.99	5.14	6.41	6.84	6.46	5.49
Basic equipment	8.42	8.17	8.54	7.94	9.26	7.66
Infection prevention	7.15	5.97	7.48	7.08	5.34	6.53
Diagnostic capacity	7.32	1.15	4.75	3.47	3.48	4.16
Essential medicines	6.62	2.35	3.98	4.69	4.94	3.82
Funding source (%)						
Public	37.0	97.1	63.0	61.8	15.0	68.9
Private	22.8	0.2	5.3	6.5	11.2	46.9
Non-state/NGO	13.0	9.0	23.0	35.9	73.6	34.7
Managing authority (%)						
Public	34.8	96.8	77.3	58.2	96.8	82.2
Private-for-profit	54.3	0.0	20.9	31.7	0.0	12.5
Private-nonprofit	10.9	3.2	1.8	10.1	3.2	5.3
Facility type (%)						
Hospital	89.1	15.2	29.0	8.4	21.9	10.7
Health centre	10.9	60.5	46.8	86.4	0.0	36.2
Health post / dispensary	0.0	24.3	24.2	5.1	32.1	53.1
Rural facilities (%)	1.1	82.1	51.6	66.5	41.3	68.1

Notes: In the table notes section, you can include some nicely aligned and formatted table notes. I kept these consistently size 9 so I didn't have to think about it.

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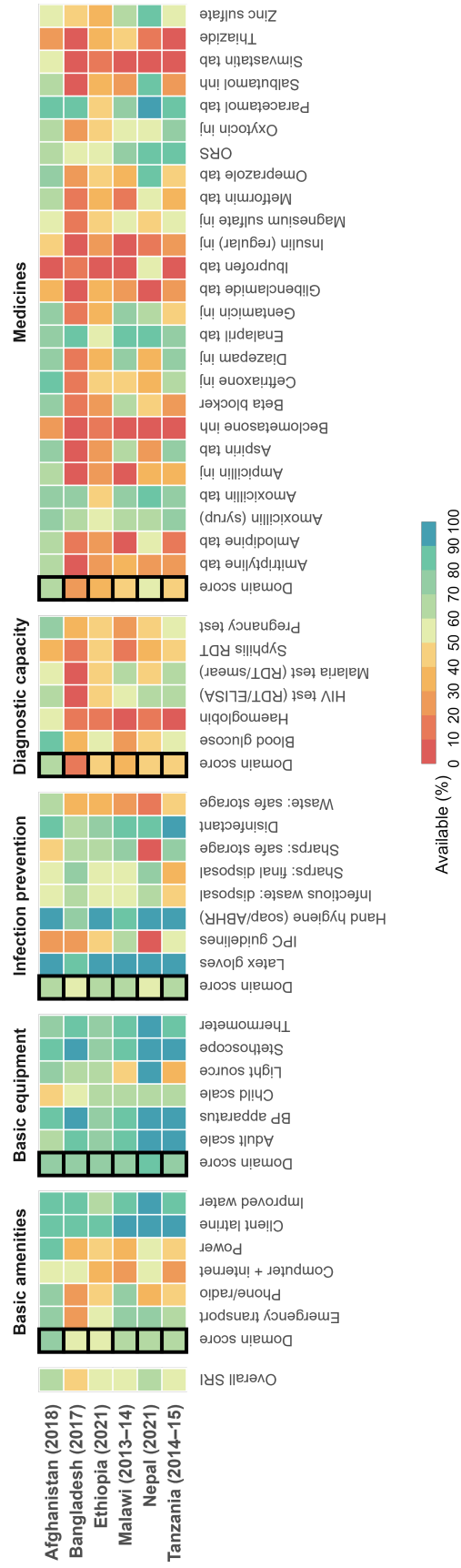


Figure 3.1: Cross-country Service Readiness Index (SRI) profiles by domain and selected tracer items. Domain scores are rescaled from 0–10 to 0–100; tracer items show the percentage of facilities with the item observed on the day of assessment. Higher values indicate greater readiness or item availability (see color bar).

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3.3 Discussion

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Chapter 4 Example Paper 3: Just Lorem Ipsum

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4.1 Methods

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4.2 Results

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4.3 Discussion

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Chapter 5 Conclusion

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5.1 Summary and synthesis of findings

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5.2 Contributions

5.2.1 *Methodological contributions*

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5.2.2 *Substantive contributions*

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5.3 Implications

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5.4 Limitations

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5.5 Future work

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5.6 Final reflections

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References

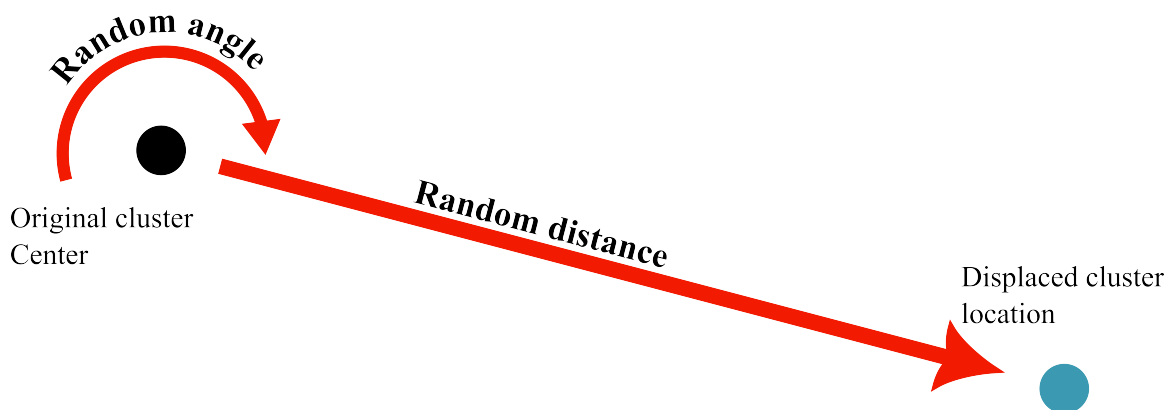
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Appendix A Appendix Example and Some Guidance

Note that appendices still use double-spacing for text. Additionally, to comply with the formatting guidance that appendices should not use table / figure numbering and should instead reference only the appendix itself, you can use the * to remove prevent number. For example
`\caption*{DHS displacement procedure visualization}`.

You can include multiple appendix files, but in this example, I have included only one.

- **Step 1:** A random angle is drawn
- **Step 2:** Random distance (between 0 km and 2 km in urban areas, between 0 km and 5 km in rural, with 1% of rural clusters displaced between 0 and 10 km)
- **Step 3:** If displaced location falls outside of admin boundary (the district level in Malawi), the steps 1 and 2 are repeated.



DHS displacement procedure visualization

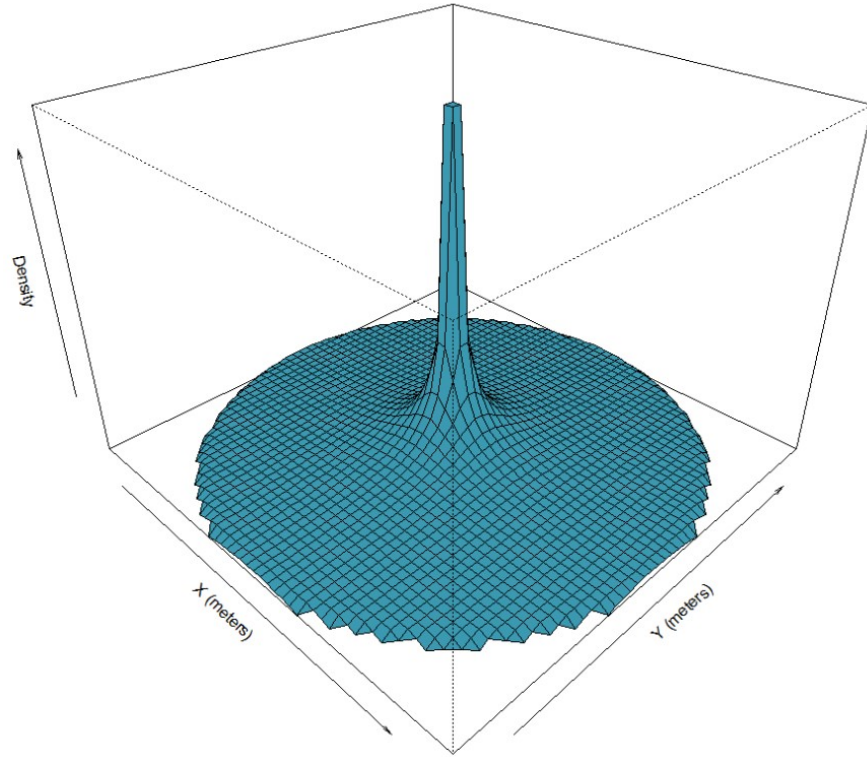
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$$f_j(x_0, y_0) = \begin{cases} \frac{1}{2\pi r_j \sqrt{x_0^2 + y_0^2}}, & \text{if } 0 < \sqrt{x_0^2 + y_0^2} \leq r_j \\ 0, & \text{otherwise.} \end{cases} \quad (\text{A.1})$$

where x_0 and y_0 denote projected coordinate offsets from the released cluster location $\tilde{s}_j$, and the singularity at $\sqrt{x_0^2 + y_0^2} = 0$ can be ignored in practice by excluding the zero-distance point from integral calculations.

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Visualization of the surface created by the equation above

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Appendix B Sensitivity tests using half-Cauchy priors for the random intercept standard deviation

Sensitivity analysis: Bayesian multilevel logistic regression models for antenatal care utilization, Malawi, using a half-Cauchy prior for the cluster-level random intercept standard deviation. Posterior median odds ratios (OR), 95% credible intervals, and posterior probability of direction (PD).

	4+ ANC visits		ANC initiation (1st trimester)	
	OR	PD [†]	OR	PD [†]
	95% CrI		95% CrI	
Health service environment (cluster-level)				
Essential medicines readiness (0–10)	1.05	99.6%	1.05	99.0%
	(1.01 – 1.08)		(1.01 – 1.10)	
Service availability (days/month)	1.00	74.1%	1.00	80.0%
	(0.99 – 1.01)		(0.99 – 1.01)	
Pr(nearest facility charges fees) (per 10pp) [§]	1.00	57.0%	0.98	100.0%
	(0.99 – 1.01)		(0.96 – 0.99)	
Distance to nearest facility (km)	0.98	98.2%	0.99	84.1%
	(0.96 – 1.00)		(0.96 – 1.01)	
Individual and household characteristics				
Rural residence	0.95	75.4%	1.16	94.8%
	(0.81 – 1.11)		(0.97 – 1.41)	
Household wealth (quintile)	1.03	97.9%	1.06	99.9%
	(1.00 – 1.07)		(1.02 – 1.10)	
Maternal age at index birth (years)	1.03	100.0%	1.02	99.7%
	(1.02 – 1.04)		(1.00 – 1.03)	
Respondent education (years)	1.03	100.0%	1.01	97.1%
	(1.01 – 1.04)		(1.00 – 1.03)	
Parity (birth order)	0.94	100.0%	0.95	99.5%
	(0.91 – 0.97)		(0.92 – 0.99)	
<i>Random intercept SD (cluster)</i>				
	0.40		0.53	
	(0.34 – 0.45)		(0.47 – 0.60)	
Observations	13,448		13,448	
Clusters (DHSID)	850		850	

ORs are posterior medians from Bayesian multilevel logistic regression models with a cluster-level random intercept (DHSID). Cluster-level health service environment measures were constructed using a probabilistic linkage procedure accounting for DHS geographic displacement. Sensitivity models used a half-Cauchy prior for the cluster-level random intercept standard deviation (see Appendix B). Credible intervals are equal-tailed (95%). [†] The posterior probability of direction (PD) is the posterior probability that the estimated effect is strictly positive or strictly negative (whichever is larger), reflecting the strength of evidence for the direction of the association. [§] Fee exposure is scaled such that a one-unit increase corresponds to a 10 percentage-point increase in the probability that the nearest facility charges fees.